

VIGNESH AMBEERU

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EDUCATION

University of South Florida, Tampa, Florida
M.S. in Computer Science

July 2023 - May 2025
GPA: 3.74/4.00

PROFESSIONAL EXPERIENCE

Accenture

Hyderabad, India

Application Development Associate

July 2021 – March 2023

- Spearheaded the migration of monolithic systems to **microservices architecture**, leveraging **Spring Boot**, **Spring Cloud Config**, and **Netflix Eureka** to enhance system scalability and reduce deployment times by **60%**. Developed scalable RESTful APIs integrating secure protocols like **Spring Security**, **JWT**, and **Google OAuth**.
- Engineered real-time data streaming architecture using **Apache Kafka** and **RabbitMQ**, while implementing **Redis caching** strategies to optimize API response times by **40%** across high-traffic services.
- Orchestrated CI/CD automation (**AWS CodePipeline**, **CodeBuild**, **CodeDeploy**) and containerization (**Docker**, **Kubernetes**) to streamline deployment workflows, reducing infrastructure setup time by **50%** and ensuring consistent environments across development stages.
- Integrated distributed tracing with **Spring Cloud Sleuth** and **Zipkin** for enhanced observability. Refined robust testing pipelines using **JUnit**, **Mockito**, and **integration tests** for Spring Boot APIs, increasing test coverage by **40%** and saving **20 hours** of manual testing weekly.
- Architected centralized logging and monitoring solution using **ELK Stack (Elasticsearch, Logstash, Kibana)**, creating real-time monitoring dashboards that reduced stakeholder reporting time by **45%** and enabled data-driven decisions.

DXC Technology

Hyderabad, India

Associate Professional Software Engineer

April 2021 – June 2021

- Designed and developed applications using **MERN (MongoDB, Express.js, React.js, and Node.js)**, enabling seamless integration across multiple business systems, ensuring security compliance.
- Implemented robust API security measures using **JWT authentication** and **middleware** in **Node.js**. Designed **role-based access control (RBAC)** and fine-tuned user permissions, reducing unauthorized access by **30%**.
- Improved front-end performance by integrating advanced **React.js** techniques, including **Code Splitting**, **Windowing**, and **Lazy Loading**, reducing load times by **35%**.
- Optimized database operations in **MongoDB** with indexing, aggregation pipelines, and schema design best practices, resulting in a **40%** improvement in query performance and faster data retrieval for high-traffic APIs.

PERSONAL PROJECTS

Uber Clone Spring Boot Project

[\[Live\]](#) | [\[Code\]](#)

Developer

October 2024 – January 2025

- Engineered a robust ride-sharing platform using **Spring Boot** and **PostgreSQL**, incorporating comprehensive user management and efficient ride-matching algorithms with **Strategy** and **Observer** Patterns. Achieved a **50%** reduction in matching time and **98%** test coverage, ensuring high system reliability.

Music Genre Virtuoso

[\[Docs\]](#) | [\[Code\]](#)

Team Lead

September 2024 – December 2024

- Collaborated with a team of three to build a custom, layered **CNN** for classifying music genres, achieving **78%** accuracy by capturing spatial features in **MFCC** representations, and achieving up to **85%** accuracy with **VGG16** model.
- Enhanced model robustness through regularization (**dropout**, **L2**), data augmentation (**pitch shifting**, **time stretching**), and deployed the system using **Flask** for real-time music genre classification.

SKILLS

Programming Languages: Java, Python, C++, C, JavaScript, TypeScript

Web Development: MERN stack, HTML, CSS, React, Express, GraphQL, Node.js, Spring Boot, Django

Databases: MySQL, PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS, Docker, Kubernetes, Kafka

Data Analytics & Visualization: Spotfire, Alteryx

Machine Learning & AI: TensorFlow, Keras, PyTorch, Scikit-learn

Project Management: Scrum, Agile Methodologies.